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Studierichting: Informatica
Oefeningenreeks 2D nr. 6.4

$$\lim_{x \rightarrow -1} f(x) = \frac{x^3 + 5x^2 + 8x + 4}{2x^3 + 4x^2 + 2x} = \frac{-1 + 5 - 8 + 4}{-2 + 4 - 2} = \frac{0}{0}$$

$$\lim_{x \rightarrow -1} \frac{(x+1)(x+2)^2}{2x(x+1)^2} = \lim_{x \rightarrow -1} \frac{(x+2)^2}{2x(x+1)} = \frac{1}{0}$$

$$\rightarrow \lim_{x \rightarrow -1+} f(x) = \frac{1}{0^-} = -\infty$$

$$\rightarrow \lim_{x \rightarrow -1-} f(x) = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0} f(x) = \frac{4}{0}$$

$$\rightarrow \lim_{x \rightarrow 0+} f(x) = \frac{4}{0^+} = +\infty$$

$$\rightarrow \lim_{x \rightarrow 0-} f(x) = \frac{4}{0^-} = -\infty$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^3}{2x^3} = \frac{1}{2}$$

References